Atoms and Molecules:

- Around 500 B.C, Indian Philosopher named Maharishi kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.

- Greek Philosphers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

→ **Laws of Chemical Combination:-**

1) **Law of Conservation of mass:-** (By Antonie Lavoiser)

- During any physical or chemical change the total mass of reactant is equal to total mass of products.

'OR'

"Mass can neither be created nor destroyed".

e.g. Mercuric oxide $\xrightarrow{heat}$ mercury + oxygen

100g 92.6g 7.4g

- Total mass of Reactants = Total Mass of Products

100g = 92.6 + 7.4 = 100g

Q) If 10g of Calcium Carbonate ($CaCO_3$) on heating gave 4.4g of carbon dioxide (CO_2) and 5.6g of Calcium Oxide (CaO). Show there observations are in agreement with law of conservation of mass?

Sol: $CaCO_3 \xrightarrow{Heat\ (\Delta)} CO_2 + CaO$

10g 4.4g + 5.6g

- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6$$

- Yes it follows Law of Conservation of mass.

- **Law of Constant Composition 'OR' Definite Proportion**

☆- Proposed by Joseph Proust.

- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g Water from any source - River, Lake, Ocean, Rain, etc. is always made up of hydrogen and Oxygen in ratio H_2O

$$2:16 = 1:8.$$

Illy. CO_2 i) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$

ii) $C + O_2 \longrightarrow CO_2$

In case (i) and (ii) CO_2 prepared by different method but elements ~~are~~ and ratio is same.

$$CO_2 \rightarrow 12:32$$
$$\Rightarrow 3:8$$

Q) When 1.375g of Cupric Oxide (CuO) was reduced on heating in a current of hydrogen the weight of copper (Cu) that remained was 1.098g. In another experiment 1.179g of copper was dissolved in nitric acid and resulting copper nitrate converted into cupric oxide by ignition. The weight of cupric oxide formed was 1.476g. Show that these results illustrate the law of ~~constran~~ constant proportion.

Page No.
Date: / /20

Sol In experiment no. 1

Mass of $CuO = 1.375g$
Mass of $Cu = 1.098g$
Mass of oxygen $= 1.375g - 1.098g = 0.277g$

$$\% \text{ of } Cu = \frac{\text{Mass of } Cu}{\text{Mass of } CuO} \times 100 = \frac{1.098 \times 100}{1.375}$$
$$= 79.85\%$$
$$= 80\%$$

$$\% \text{ of Oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of } CuO} \times 100 = \frac{0.277 \times 100}{1.375}$$
$$= 20\%$$

In experiment no. 2

Mass of $CuO = 1.476$
Mass of $Cu = 1.179$
Mass of oxygen $= 1.476 - 1.179 = 0.297$

$$\% \text{ of } Cu = \frac{\text{Mass of } Cu}{\text{Mass of } CuO} = \frac{1.179 \times 100}{1.476}$$
$$= 79.8\%$$
$$= 80\%$$

$$\% \text{ of oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100$$

$$= \frac{0.297}{1.476} \times 100$$

$$= 20\%$$

$\therefore$ Law of constant composition followed.

⇒ **Dalton's Atomic Theory:-**

- Proposed by John Dalton.

- John Dalton studied structure of matter and proposed following points:-

1) Matter is made up of extremely small particles called as Atoms.

2) Atoms of an element are similar in every respect, i.e; Mass, shape, size, etc.

3) Atoms of different elements are different in every respect i.e; mass, size, shape etc.

4) Atoms is the smallest particle that takes part in a chemical reaction.

5) Atoms combine together in simple whole no. ratio to form molecules.

6) Atoms are indivisible i.e; they can't be divided further.

⇒ Limitations of Dalton's Atomic theory:-

1) This theory did not explain why the atoms of different elements are different i.e; this theory did not talk about internal structure of Atom.

2) Why and how atoms combine and form molecules

3) Atom is indivisible according to this theory but Atom can be divided.

⇒ **Atom**:- It is the smallest particle that may or may not have independent existence.

e.g. Atoms that exist independently = Na, C, K, B, etc.

Atoms that exist in combined form = O_2, H_2, N_2, etc.

⇒ **Symbols of Atoms of different elements**:-

- ~~Joh~~ John Dalton proposed some nortification to represent symbols of Atoms.

Atoms	symbols.
e.g.: Carbon →	●
Oxygen →	○
Hydrogen →	⊙

- These symbols were later discarded.

⇒ Berzelius proposed that the symbols should include letters from name of atom.

- IUPAS - International Union of pure and applied chemistry; developed following symbols:

A) Elements that use first letter of name as symbol:

i) Hydrogen - H
ii) Oxygen - O
iii) Nitrogen - N
iv) Carbon - C, etc.

B) First two letters in symbols:

Aluminium - Al
Helium - He
Lithium - Li
Silicon - Si
Neon - Ne
Argon - Ar, etc.

c) First and third letter of name:-

Magnesium - Mg
Chlorine - Cl
zinc - zn , etc.

d) Symbols from Latin Names:-

Iron - ~~Fers~~ Ferrum - Fe
Lead - Plumbum - Pb
Gold - Aurum - Au
Sodium - Natrium - Na
Potassium - kalium - K
Tin - stannum - Sn

☆ ⇒ Atomic Mass:-

- In Ancient times isolation of ~~som~~ single atom and calculation of its mass was difficult.
- Mass of any atom was calculated with respect to some standard or refrence value. So the

mass calculated is called as Relative Atomic mass.

→ Relative Atomic mass calculated with respect to some standard value.

- Scale on which these masses were expressed is called as Atomic Mass scale.

→ The refrence chosen for calculation of relative atomic mass.

1. Hydrogen
But mass of some atoms came out to be fractional so it was discarded.

2. Oxygen
But mass of some atoms showed variation using oxygen as standard so it was also discarded.

3. Finally IUPAS set carbon as standard and

mass of other atoms was calculated using carbon as 12 units on atomic mass scale.

⇒ **Atomic Mass:-** It is defined as relative atomic mass of an atom with respect to C-12 taken as 12 units.

⇒ **Unit:-** amu → Atomic mass unit

e.g. For Hydrogen - 1 amu. Oxygen = 8 amu.

⇒ **Atomic Mass Unit - (amu):-**

- Atomic mass unit is defined as $\frac{1}{12^{th}}$ mass of carbon atom.

$$1 amu = 1.66 \times 10^{-24} g$$

⇒ **Molecules:-**

- Molecules is defined as the smallest particle of matter that exists freely, the properties of that substance. Individual properties of atom are not present in molecule.

e.g. CO_2, H_2O, NH_3, etc.

Molecules ~~of~~ - 2 Types

Molecule of Element. | Molecule of compound

1) Molecule of Element:-

- Also called as ~~Homot~~ HOMOATOMIC Molecules.
- These are the molecules that are made up of same type of atoms.

e.g. O_2, O_3, P_4 etc.

2) Molecule of Compound:-

- Also called as HETEROATOMIC molecules.
- These are the molecules that are made up of

different type of atoms.

e.g CO_2, H_2O, NH_3, etc.

⇒ <u>Atomicity</u>:- It is the no. of atoms present in a molecule.

Diatomic → 2 atoms
Triatomic → 3 atoms
Tetraatomic → 4 atoms
Pentaatomic → 5 atoms.

(E.g) Homoatomic	Heteroatomic
• Monoatomic - He, Ne, Ar, Xe, Noble gases.	• <u>Diatomic</u>: NaCl, KCl, CO
	• <u>Triatomic</u>: H_2O, CO_2
• <u>Diatomic</u>:- O_2, H_2, N_2	• <u>Tetraatomic</u>:- NH_3
• <u>Triatomic</u>: O_3	
• <u>Tetraatomic</u>: P_4	